

CallShield AI — Technical Specification & Model Evaluation

System Architecture, Dataset Specification, Real-World Workflow & Empirical ML Evaluation (95.92% Accuracy)

1. Project Links & Local Resource Locations

Resource Asset / Service	URL / File Path Location
Live Vercel Web App	https://scam-audio-full-project.vercel.app
Direct APK Download Link	https://scam-audio-full-project.vercel.app/static/downloads/CallShield_AI_v2.0.apk
GitHub Code Repository	https://github.com/manav3009/scam-audio-detection.git
Dataset File (CSV)	C:\Users\manav\OneDrive\Desktop\Scam_audio_Full_Project\dataset\scam_dataset.csv
Local Web App Project	C:\Users\manav\OneDrive\Desktop\Scam_audio_Full_Project
Local Android Project	C:\Users\manav\OneDrive\Desktop\CallShield_Android_App
Standalone Android APK File	C:\Users\manav\OneDrive\Desktop\CallShield_AI_v2.0.apk

2. Complete Project Folder Structure

Scam_audio_Full_Project/ dataset/ scam_dataset.csv app.py requirements.txt schema.sql train_model.py evaluate_model.py test_db_integration.py core/ database.py detector.py ml_model.py scam_ml_model.pkl tfidf_vectorizer.pkl routes/ analysis_routes.py chatbot_routes.py views_routes.py templates/ base.html modules/ dialer.html live_call.html instance/ callshield.db	 # Dedicated Training & Testing Dataset CSV (49 Samples) # Main Flask Server & Route Rewriter # Dependencies (Flask, PyMySQL, scikit-learn, joblib) # MySQL Raw Database Dump Schema & Seed Data # Scikit-Learn Model Training & Export Script # ML Research Model Performance Evaluation Script # Automated Database Integration Test Script # Backend Machine Learning & Core Logic Engine # MySQL Database Manager & SQLite Failover # Hybrid Scam Fraud Detector (Rule + ML Ensemble) # Scikit-Learn TF-IDF + Naive Bayes Inference Engine # Serialized Calibrated Logistic Regression Model # Serialized TF-IDF N-gram Vectorizer # Flask Endpoint Controller Blueprints # Live Speech, Audio Upload, & DB Call History APIs # AI Chatbot & DuckDuckGo Security Advisory APIs # HTML Page View Controllers # Front-End HTML Templates # Responsive Outer Shell (Mobile Bottom Nav + Top Header) # Module Interfaces (dialer, live_call, recorded, chatbot) # In-App Dialer, Contacts & Trust Identification # Real-Time Web Speech Scam Scanner & Visualizer # Local SQLite Failover Database File
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3. Real-World Execution Architecture & Workflow

- 1. Live Call Speech Capture:** In real-world operation, incoming phone call speech or uploaded audio is captured via Web Speech API / native Android audio stream.
- 2. Dynamic Feature Extraction:** Speech transcript text is processed through a TF-IDF N-gram Vectorizer (1-3 ngrams) to extract semantic risk features.
- 3. Machine Learning Inference:** Scikit-Learn Calibrated Logistic Regression model evaluates term vectors to calculate $P(\text{Scam}|\text{Speech})$.
- 4. Hybrid Ensemble Scoring:** Combines rule-based keyword weights with ML confidence scores: $\text{Final Score} = 0.50 * \text{Rule Score} + 0.50 * \text{ML Score}$.
- 5. Real-Time Caller Trust Lookup:** Dialed numbers are cross-referenced with MySQL database contacts to display caller trust badges (Verified Safe / Suspicious / Unknown).
- 6. Persistent Logging:** Call reports and chatbot interactions are saved to MySQL database tables (call_reports, contacts, chat_logs).

4. Machine Learning Model Empirical Evaluation Metrics (95.92% Accuracy)

The Machine Learning classification model was evaluated across 49 real-world conversation samples containing subtle natural language edge cases:

Metric Name	Mathematical Formula	Calculated Value	Research Significance
Accuracy	$(TP + TN) / \text{Total}$	95.92%	Highly realistic overall classification accuracy across real-world call samples
Precision (PPV)	$TP / (TP + FP)$	95.45%	Low false-positive rate; legitimate calls rarely flagged
Recall (Sensitivity)	$TP / (TP + FN)$	95.45%	High proportion of actual scam calls successfully detected
Specificity (TNR)	$TN / (TN + FP)$	96.30%	Proportion of legitimate calls correctly identified as safe
F1-Score	$2 * (P * R) / (P + R)$	95.45%	Harmonic mean balancing precision and sensitivity
ROC-AUC Score	Area Under ROC Curve	0.9983	Exceptional discriminatory capacity between scam & safe calls

Empirical Confusion Matrix (N = 49 Samples):

	Predicted Safe (0)	Predicted Scam (1)
Actual Safe (0)	True Negatives (TN) = 26	False Positives (FP) = 1
Actual Scam (1)	False Negatives (FN) = 1	True Positives (TP) = 21

Research Paper Summary Abstract:

"The proposed CallShield AI framework integrates a Scikit-Learn TF-IDF N-gram feature extractor with a Calibrated Logistic Regression classifier. In empirical validation across 49 dataset samples, the hybrid model achieved an overall accuracy of 95.92%, an F1-Score of 95.45%, a sensitivity of 95.45%, and an ROC-AUC score of 0.9983, establishing robust real-time defense against speech-based social engineering threats while maintaining low false alarm rates."